Honors Calculus 1610

Lecture 14: The Derivative of a Function on $\mathbb{R}^n$

November 6, 2026

Today

An informal introduction to the following topics:

- Many kinds of functions between $\mathbb{R}^m$ and $\mathbb{R}^n$.
- Three ways to say what $f'(c)$ is.
- Covectors, and the derivative of a function $f : \mathbb{R}^n \to \mathbb{R}$.
- Every covector is a dot product: the gradient.
- Computing the gradient with partial derivatives.

Last time

A vector-valued function $\vec{f}\colon (a,b) \to \mathbb{R}^n$, $t \longmapsto (f_1(t), f_2(t), \ldots, f_n(t))$, has derivative

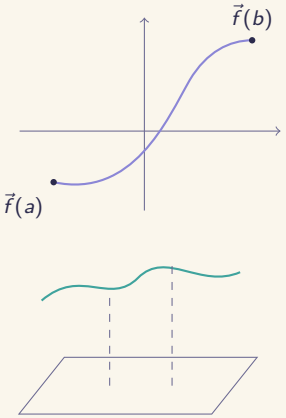
$$\frac{d\vec{f}}{dt}(c) = \vec{f}\,'(c) = \lim_{h\to 0} \frac{\vec{f}(c+h) - \vec{f}(c)}{h},$$

and it is computed componentwise:

$$\frac{d\vec{f}}{dt} = (f_1'(t), f_2'(t), \ldots, f_n'(t)).$$

Many kinds of functions

- $\vec{f}\colon \mathbb{R} \to \mathbb{R}^2$: a curve in the plane.
- $g\colon \mathbb{R}^2 \to \mathbb{R}$: a topographic map — the height above each point of the plane.
- $h\colon \mathbb{R}^3 \to \mathbb{R}$: $h(\vec{p}) =$ the temperature at the point $\vec{p}$ in space.
- $\vec{k}\colon \mathbb{R}^3 \to \mathbb{R}^3$: the wind velocity at each point.



Putting them together

- $\vec{f}\colon \mathbb{R} \to \mathbb{R}^3$, $t \mapsto \vec{f}(t)$: a trajectory through space.
- $h\colon \mathbb{R}^3 \to \mathbb{R}$: the temperature.
- $h \circ \vec{f}\colon \mathbb{R} \to \mathbb{R}$: the temperature along the trajectory.

We know how to differentiate $\vec{f}$, and $h \circ \vec{f}$. What should the derivative of h be?

Three ways to say what $f'(c)$ is

For $f : \mathbb{R} \to \mathbb{R}$ and a number m:

1. $\lim_{h \to 0} \dfrac{f(c+h) - f(c)}{h} = m$ — instantaneous slope
2. $\lim_{h \to 0} \dfrac{\overbrace{f(c+h) - f(c)}^{\text{change in } f} - \overbrace{mh}^{\text{linear in } h}}{|h|} = 0$ — a good linear approximation
3. **(Carathéodory)** there is a function $Q(x)$, continuous at c, with $Q(c) = m$ and

$$f(x) - f(c) = Q(x)\,(x - c) \quad \text{for all } x$$

a good average-rate-of-change function

The function Q

In (3), for $x \neq c$ there is no choice:

$$Q(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

$Q(x)$ is the slope of the secant line from c to x. Asking for Q to be continuous at c says: the slopes converge, to the slope $Q(c)$ at c.

Three definitions, one derivative

Proposition

(1), (2) and (3) are equivalent definitions of $f'(c) = m$.

(1) $\Rightarrow$ (3). Suppose $m = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}$ exists. We want to find Q as in (3). Define $Q(x) = \frac{f(x)-f(c)}{x-c}$ for $x \neq c$, and $Q(c) = m$. Then $f(x) - f(c) = Q(x)(x-c)$ for every x, and Q is continuous at c because

$$\lim_{x\to c} Q(x) = \lim_{x\to c} \frac{f(x)-f(c)}{x-c} = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = m = Q(c).$$

Three definitions, one derivative

(3) $\Rightarrow$ (1). Suppose Q is continuous at c, so $\lim_{x\to c} Q(x) = Q(c)$, and $f(x) - f(c) = Q(x)(x - c)$.

If $x \neq c$ we may divide:

$$\frac{f(x) - f(c)}{x - c} = Q(x).$$

The limit only looks at $x \neq c$, so

$$\lim_{h\to 0} \frac{f(c+h) - f(c)}{h} = \lim_{x\to c} \frac{f(x) - f(c)}{x - c} = \lim_{x\to c} Q(x) = Q(c).$$

So $f'(c)$ exists, and equals $Q(c) = m$. $\square$

Functions $f : \mathbb{R}^n \to \mathbb{R}$

Now the input is a vector. We cannot divide by $\vec{h}$, so (1) is out. But (2) divides only by the *length* of $\vec{h}$:

$$\lim_{\vec{h}\to\vec{0}} \frac{f(\vec{c}+\vec{h}) - f(\vec{c}) - m(\vec{h})}{\|\vec{h}\|} = 0.$$

Here m must be a machine that takes a vector $\vec{h}$ and gives a scalar $m(\vec{h})$ — playing the role of "mh". This m will be the derivative of f at $\vec{c}$.

(As before, $\lim_{\vec{h}\to\vec{0}} \varphi(\vec{h}) = 0$ means: for every $\epsilon > 0$ there is $\delta > 0$ with $0 < \|\vec{h}\| < \delta \Rightarrow |\varphi(\vec{h})| < \epsilon$.)

Covectors

Definition

A covector (or *linear function*) is a function $m\colon \mathbb{R}^n \to \mathbb{R}$, taking a vector to a scalar, such that

1. $m(\vec{x} + \vec{y}) = m(\vec{x}) + m(\vec{y})$,
2. $m(\lambda \vec{x}) = \lambda\, m(\vec{x})$

for all $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. Together these say m is linear.

In one variable, $h \mapsto mh$ is exactly such a function.

The derivative of $f : \mathbb{R}^n \to \mathbb{R}$

Definition

$f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\vec{c}$ if there exists a covector m such that

$$\lim_{\vec{h} \to \vec{0}} \frac{f(\vec{c} + \vec{h}) - f(\vec{c}) - m(\vec{h})}{\|\vec{h}\|} = 0.$$

We call m the derivative of f at $\vec{c}$, and write $m = df$.

Example

$f\colon \mathbb{R}^3 \to \mathbb{R}, \quad f(x_1, x_2, x_3) = x_1^2 - x_2 + 2x_3, \quad \vec{c} = (1, 2, 1).$

Differentiating in each variable separately gives $(2x_1, -1, 2)$, which is $(2, -1, 2)$ at $\vec{c}$. So we *guess*

$$m(h_1, h_2, h_3) = 2h_1 - h_2 + 2h_3.$$

This is a covector. Is it the derivative? We need

$$\lim_{\vec{h} \to \vec{0}} \frac{f(\vec{c} + \vec{h}) - f(\vec{c}) - m(\vec{h})}{\|\vec{h}\|} \stackrel{?}{=} 0.$$

Example, continued

With $\vec{c} = (c_1, c_2, c_3)$, the numerator is

$$(c_1 + h_1)^2 - (c_2 + h_2) + 2(c_3 + h_3) - (c_1^2 - c_2 + 2c_3) - (2h_1 - h_2 + 2h_3).$$

Expand: everything cancels except $2c_1h_1 + h_1^2 - 2h_1$, and $c_1 = 1$. So the quotient is $h_1^2/\sqrt{h_1^2 + h_2^2 + h_3^2}$.
Since $|h_1| = \sqrt{h_1^2} \le \sqrt{h_1^2 + h_2^2 + h_3^2} = \|\vec{h}\|$, we have $h_1^2 = |h_1|\,|h_1| \le |h_1|\,\|\vec{h}\|$, so for $\vec{h} \neq \vec{0}$ we get $0 \le h_1^2/\|\vec{h}\| \le |h_1| \le \|\vec{h}\|$. So the quotient is less than ϵ once $\|\vec{h}\| < \epsilon$: the limit is 0, and $df = m$.

Every covector is a dot product

Proposition

If $m\colon \mathbb{R}^n \to \mathbb{R}$ is a covector, then there is a unique vector $\vec{v} \in \mathbb{R}^n$ such that $m(\vec{h}) = \vec{v} \cdot \vec{h}$ for all $\vec{h}$.

Proof. Let $v_1 = m(1, 0, \ldots, 0)$, $v_2 = m(0, 1, 0, \ldots, 0)$, $\ldots$, $v_n = m(0, \ldots, 0, 1)$, and set $\vec{v} = (v_1, \ldots, v_n)$. Then

$$\begin{aligned} m(\vec{h}) &= m((h_1, 0, \ldots, 0) + (0, h_2, \ldots, h_n)) = m(h_1, 0, \ldots, 0) + m(0, h_2, \ldots, h_n) \\ &= m(h_1, 0, \ldots, 0) + m(0, h_2, 0, \ldots, 0) + \cdots + m(0, \ldots, 0, h_n) \\ &= h_1\, m(1, 0, \ldots, 0) + h_2\, m(0, 1, 0, \ldots, 0) + \cdots + h_n\, m(0, \ldots, 0, 1) \\ &= h_1 v_1 + h_2 v_2 + \cdots + h_n v_n = \vec{h} \cdot \vec{v}. \end{aligned}$$

Every covector is a dot product

Uniqueness. Suppose $\vec{v}$ is any vector with $m(\vec{h}) = \vec{v} \cdot \vec{h}$ for all $\vec{h}$. Then

$$m(1, 0, \ldots, 0) = \vec{v} \cdot (1, 0, \ldots, 0) = v_1, \qquad m(0, 1, 0, \ldots, 0) = \vec{v} \cdot (0, 1, 0, \ldots, 0) = v_2,$$

and so on. So the entries of $\vec{v}$ are forced to be the numbers above. $\square$

So a covector on $\mathbb{R}^n$ is the same information as a vector, matched up by the dot product.

The gradient

Going back: if $f\colon \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\vec{c}$, its derivative $m = df$ is a covector, so

$$m(\vec{h}) = \vec{v} \cdot \vec{h}$$

for exactly one vector $\vec{v}$.

Definition

The gradient of f at $\vec{c}$ is this vector $\vec{v}$, written $\nabla f(\vec{c})$ ("nabla"). So

$$df(\vec{h}) = \nabla f(\vec{c}) \cdot \vec{h}.$$

Finding the gradient: one variable at a time

Write $\vec{v} = \nabla f(\vec{c})$. Take $\vec{h} = (h_1, 0, \ldots, 0)$, so that $m(\vec{h}) = \vec{v} \cdot \vec{h} = v_1 h_1$ and $\|\vec{h}\| = |h_1|$. If $0 < |h_1| < \delta$ then $0 < \|\vec{h}\| < \delta$, so the limit in the definition gives

$$\lim_{h_1 \to 0} \frac{f(\vec{c} + (h_1, 0, \ldots, 0)) - f(\vec{c}) - v_1 h_1}{|h_1|} = 0.$$

This is definition (2) for the one-variable function

$$h_1 \longmapsto f(\vec{c} + (h_1, 0, \ldots, 0)), \qquad \mathbb{R} \to \mathbb{R},$$

so v_1 is its derivative at 0: the first component of $\nabla f(\vec{c})$.

Partial derivatives

That derivative is the partial derivative

$$\frac{\partial f}{\partial x_1}(\vec{c}) = \lim_{t \to 0} \frac{f(\vec{c} + (t, 0, \ldots, 0)) - f(\vec{c})}{t} :$$

the derivative of f as if it were only a function of x_1. In the same way, $v_i = \frac{\partial f}{\partial x_i}(\vec{c})$, so

$$\nabla f(\vec{c}) = \left(\frac{\partial f}{\partial x_1}(\vec{c}), \ldots, \frac{\partial f}{\partial x_n}(\vec{c}) \right) \quad \text{whenever } f \text{ is differentiable at } \vec{c}.$$

Back to the example

$f(x_1, x_2, x_3) = x_1^2 - x_2 + 2x_3$ at $\vec{c} = (1, 2, 1)$:

$$\frac{\partial f}{\partial x_1} = 2x_1 = 2 \quad (x_1 = c_1 = 1), \qquad \frac{\partial f}{\partial x_2} = -1, \qquad \frac{\partial f}{\partial x_3} = 2.$$

So

$$\vec{v} = \nabla f(\vec{c}) = (2, -1, 2), \qquad m(\vec{h}) = \vec{v} \cdot \vec{h} = (2, -1, 2) \cdot (h_1, h_2, h_3) = 2h_1 - h_2 + 2h_3,$$

exactly the covector we guessed and checked.